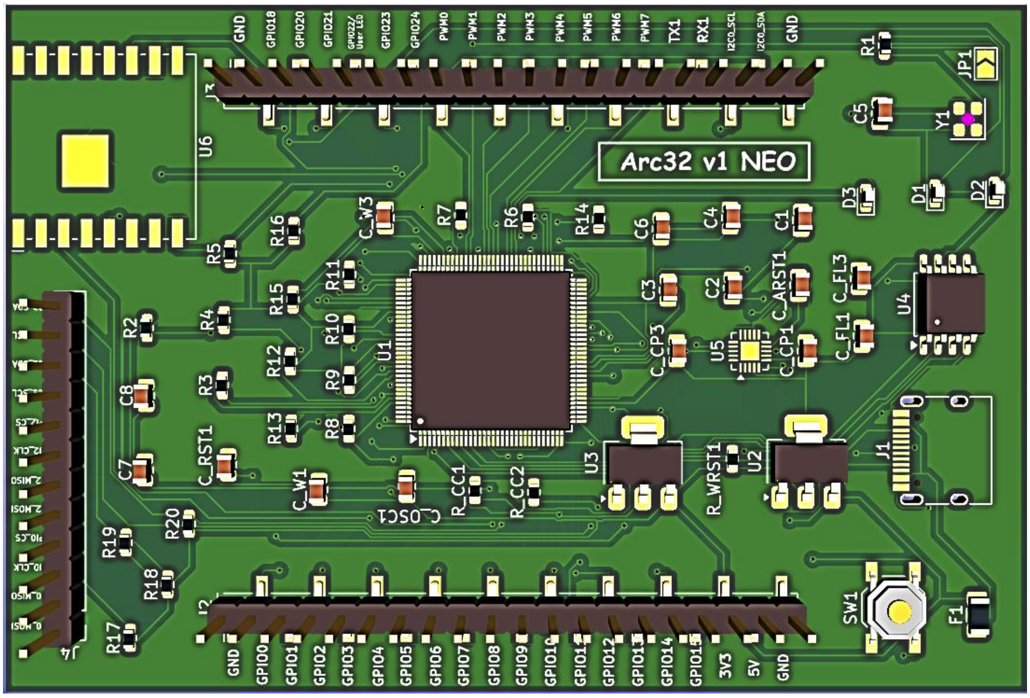
Arc32 v1

THEJAS32 RISC-V (RV32IM) Arduino-Compatible Development Board



Arc32 v1 NEO — top-side 3D render (KiCad)

MCU	THEJAS32 SoC — VEGA ET1031 32-bit RISC-V (RV32IM), 100 MHz
Memory	256 KB embedded SRAM + 2 MB external SPI NOR flash (W25Q16JVSS)
Wireless	Onboard BW16 (RTL8720DN) Wi-Fi 4 + Bluetooth LE 5 module
USB	USB-C, CP2102N USB-UART bridge for programming / power / serial
I/O	24 usable GPIO, 8 PWM, 3 UART, 4 SPI, 3 I2C (2 SPI + 2 I2C + 1 UART on headers)
Software	Programmable from the Arduino IDE via a custom Boards Manager package
Board size	60.4 mm x 89.8 mm

Designed & maintained by Nishil Patel | github.com/nishil-26/arc32-sdk | Rev 1.0 — July 2026

1. Description

The Arc32 v1 NEO is an Arduino-form-factor development board built around the **THEJAS32** SoC — an indigenous 32-bit RISC-V (RV32IM) microcontroller developed by the Centre for Development of Advanced Computing (C-DAC) under India's Digital India RISC-V (DIR-V) programme, using the VEGA ET1031 CPU core. The board pairs the THEJAS32 with a USB-C interface, an onboard SPI NOR flash for standalone boot, and a BW16 (RTL8720DN) Wi-Fi/Bluetooth LE module, and exposes the SoC's GPIO, PWM, UART, SPI and I2C peripherals on three 0.1" pin headers (J2, J3, J4).

The Arc32 SDK (a custom Arduino core + Boards Manager package, **github.com/nishil-26/arc32-sdk**) allows the board to be programmed directly from the Arduino IDE, in the same workflow used for Arduino AVR, ESP32 or SAMD boards — the IDE downloads the RISC-V GCC toolchain automatically and sketches are uploaded over the USB-C port using the XMODEM protocol.

This document describes the Arc32 v1 NEO board only. For the full electrical and architectural reference of the THEJAS32 SoC itself, see the official THEJAS32 Datasheet (C-DAC / VEGA Processors, Doc Ver 1.0).

1.1 Key Features

- VEGA ET1031 32-bit RISC-V (RV32IM) CPU, 100 MHz, 1.3 DMIPS/MHz, hardware multiply/divide
- 256 KB internal SRAM; 2 MB (16 Mbit) external SPI NOR flash for standalone program boot
- Onboard BW16 (RTL8720DN) module for Wi-Fi 4 (2.4 GHz) and Bluetooth LE 5, bridged to the SoC over a dedicated SPI + UART + I2C link
- USB-C connector with CP2102N USB-to-UART bridge for power, programming and serial monitor — no external programmer required
- 24 general-purpose I/O pins broken out on Arduino-style headers, plus 12 additional signal pins (2x SPI, 2x I2C) on an auxiliary header
- 8 hardware PWM channels, 3 UART, 4 SPI and 3 I2C peripheral instances on the SoC
- One-button reset, dedicated boot-mode jumper (UART / SPI-flash boot), Power / Heartbeat / User LEDs, resettable 500 mA polyfuse on the USB rail
- Fully supported by the free, open-source Arc32 Arduino core (digitalWrite/Read, analogWrite (PWM), Serial, Wire, SPI, millis/micros, attachInterrupt, ...)

1.2 Board Block Diagram

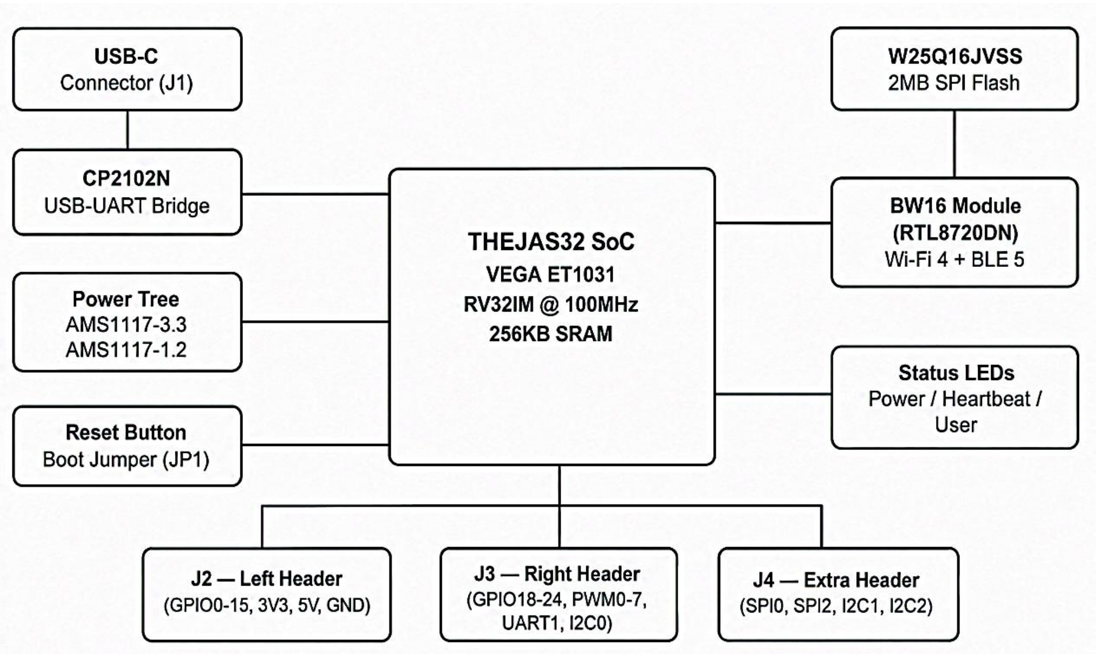


Figure 1. Arc32 v1 NEO block diagram

2. Pinout

All three headers use a 2.54 mm (0.1") pitch, Arduino/ESP32-style single-row layout. J2 and J3 also carry the Arduino digital-pin numbers (D0-D29) used by the Arc32 Arduino core.

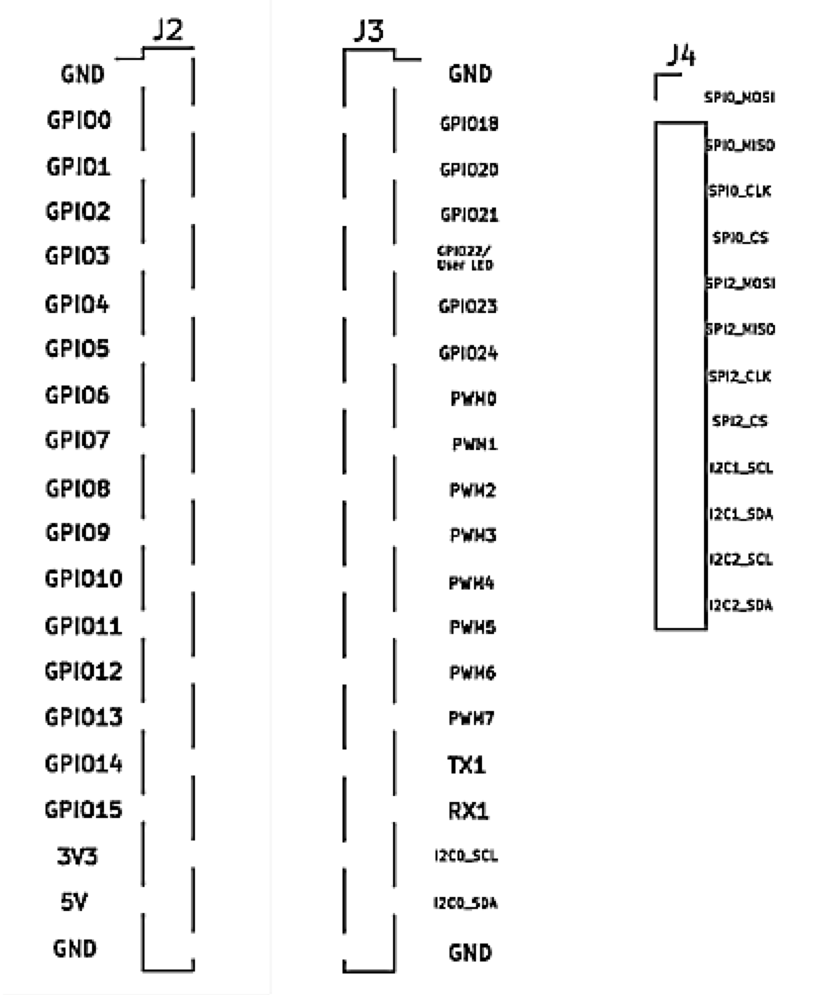


Figure 2. Header locations (J2 left, J3 right, J4 top / auxiliary)

2.1 J2 — Left Header (20-pin)

Pin	Signal	THEJAS32 pin/notes
1	GND	Ground
2	D0 / GPIO0	THEJAS32 pin 74
3	D1 / GPIO1	THEJAS32 pin 73
4	D2 / GPIO2	THEJAS32 pin 72
5	D3 / GPIO3	THEJAS32 pin 71
6	D4 / GPIO4	THEJAS32 pin 66
7	D5 / GPIO5	THEJAS32 pin 65
8	D6 / GPIO6	THEJAS32 pin 64
9	D7 / GPIO7	THEJAS32 pin 63
10	D8 / GPIO8	THEJAS32 pin 62
11	D9 / GPIO9	THEJAS32 pin 61
12	D10 / GPIO10	THEJAS32 pin 58

13	D11 / GPIO11	THEJAS32 pin 55, IRQ-capable
14	D12 / GPIO12	THEJAS32 pin 54, IRQ-capable
15	D13 / GPIO13	THEJAS32 pin 53, IRQ-capable
16	D14 / GPIO14	THEJAS32 pin 52, IRQ-capable
17	D15 / GPIO15	THEJAS32 pin 51, IRQ-capable
18	3V3	3.3V regulated output (from AMS1117-3.3)
19	5V	5V rail (from USB-C VBUS, fused)
20	GND	Ground

2.2 J3 — Right Header (20-pin)

Pin	Signal	THEJAS32 pin / notes
1	GND	Ground
2	D16 / GPIO18	THEJAS32 pin 2
3	D17 / GPIO20	THEJAS32 pin 128
4	D18 / GPIO21	THEJAS32 pin 127
5	D19 / GPIO22	THEJAS32 pin 122 — shared with onboard User LED (D3)
6	D20 / GPIO23	THEJAS32 pin 121
7	D21 / GPIO24	THEJAS32 pin 120
8	D22 / PWM0	THEJAS32 pin 86
9	D23 / PWM1	THEJAS32 pin 85
10	D24 / PWM2	THEJAS32 pin 82
11	D25 / PWM3	THEJAS32 pin 81
12	D26 / PWM4	THEJAS32 pin 80
13	D27 / PWM5	THEJAS32 pin 77
14	D28 / PWM6	THEJAS32 pin 76
15	PWM7	THEJAS32 pin 75 — not pre-assigned in the Arduino core
16	TX1 (UART1_TX)	THEJAS32 pin 45
17	RX1 (UART1_RX)	THEJAS32 pin 50
18	SCL (I2C0_SCL)	THEJAS32 pin 33
19	SDA (I2C0_SDA)	THEJAS32 pin 34
20	GND	Ground

2.3 J4 — Extra Header (12-pin)

J4 breaks out the SoC peripherals not already present on J2/J3: the SPI0 and SPI2 controllers, and the I2C1 and I2C2 controllers.

Pin	Signal	THEJAS32 pin / notes
1	SPI0_MOSI	THEJAS32 pin 87
2	SPI0_MISO	THEJAS32 pin 90
3	SPI0_CLK	THEJAS32 pin 91
4	SPI0_CS	THEJAS32 pin 92
5	SPI2_MOSI	THEJAS32 pin 97
6	SPI2_MISO	THEJAS32 pin 98
7	SPI2_CLK	THEJAS32 pin 101
8	SPI2_CS	THEJAS32 pin 102
9	I2C1_SCL	THEJAS32 pin 96
10	I2C1_SDA	THEJAS32 pin 95
11	I2C2_SCL	THEJAS32 pin 32
12	I2C2_SDA	THEJAS32 pin 31

2.4 Reserved / Not Broken Out

A few THEJAS32 signals are used internally by the board and are not available on any header:

- GPIO16, GPIO17 — control the BW16 module's EN / WIFI_RST lines (reserved for the onboard Wi-Fi/BT module; do not repurpose)
- GPIO25-31 — hard-strapped through 4.7 kΩ resistors to set the default UART0 boot baud-rate divider (per THEJAS32 boot-mode requirements); not general-purpose I/O
- GPIO19 — not routed to a header pin
- SPI3, UART0 and UART2 — reserved for the onboard SPI flash, USB-UART bridge, and BW16 log/debug UART respectively

2.5 Arduino Pin Mapping (Arc32 core, pins_arduino.h)

The Arc32 Arduino core exposes digital pins D0-D29 as shown below. PWM-capable pins (D23-D29) support analogWrite(); GPIO0-GPIO11 (D0-D11) support attachInterrupt().

Arduino name	SoC signal	Header	Notes
D0-D15	GPIO0-GPIO15	J2, pins 2-17	digitalRead/Write; D0-D11 interrupt-capable
D16	GPIO18	J3, pin 2	digitalRead/Write
D17-D21	GPIO20-GPIO24	J3, pins 3-7	digitalRead/Write; D19=GPIO22=onboard User LED
D22-D28	PWM0-PWM6	J3, pins 8-14	analogWrite() PWM output
D29	not mapped	—	PWM7 (pin 75) is wired to J3 but unused by default
LED_BUILTIN	D19 (GPIO22)	onboard	Blue User LED (D3)
Serial	UART0	USB-C (J1)	via CP2102N — primary programming/debug port
Serial1	UART1	J3 TX1/RX1	PIN_SERIAL1_TX=45, PIN_SERIAL1_RX=50

Arduino name	SoC signal	Header	Notes
SPI	SPI0	J4, pins 1-4	default SPI object
Wire	I2C0	J3 SCL/SDA	hardware pins 33/34

Source: variants/arc32/pins_arduino.h, arc32-sdk repository.

3. Onboard Peripherals

3.1 USB-C & Programming Interface (J1, U5)

A USB-C receptacle (J1) connects to a Silicon Labs CP2102N USB-to-UART bridge (U5), which drives THEJAS32 UART0 (RX0/TX0) and the PUSH_RESETN reset line. This is the board's sole USB connection — it provides 5V power, the serial programming path (XMODEM over UART0), and the Serial Monitor link, all through one cable.

3.2 Onboard SPI Flash (U4)

A Winbond W25Q16JVSS (16 Mbit / 2 MB) SPI NOR flash is connected to THEJAS32's SPI3 bus. When the board is set to SPI-flash boot mode (JP1 shorted), the bootloader copies the first 250 KB from this flash into SRAM at power-up and jumps to the program, letting the board run standalone without a PC connection.

3.3 Onboard Wireless — BW16 Module (U6)

A BW16 module (Realtek RTL8720DN-based, providing 2.4 GHz Wi-Fi 4 and Bluetooth LE 5) is fitted as a companion wireless co-processor. It connects to the THEJAS32 over a dedicated SPI1 link, a UART log/debug channel, an I2C link, and GPIO16/GPIO17 (EN / WIFI_RST). These signals are internal to the board and are not exposed on J2/J3/J4.

Wi-Fi/BLE support is not yet exposed through a ready-made Arduino library in the current SDK release — driver work for the BW16 link is in progress.

3.4 Reset, Boot Jumper & Fuse

Ref	Function	Description
SW1	Reset button	Pulls PUSH_RESETN low; simple RC reset network (R4 10kΩ / C_RST1 100nF), matching the THEJAS32 datasheet's no-reset-IC reference circuit.
JP1	Boot Jumper	Selects BOOT_SEL. Open (default) = UART/XMODEM boot mode. Shorted = SPI-flash standalone boot mode.
F1	500 mA polyfuse	Resettable fuse on the USB-C VBUS (5V) rail, protecting the host port from overcurrent.

3.5 Status LEDs

Ref	Name	Description
D1	Power LED	Lit whenever 3.3V is present
D2	Heartbeat LED	Driven by THEJAS32 PROC_HB (pin 16) — a hardware CPU-alive heartbeat signal
D3	User LED	Driven by GPIO22 (D19) — the Arduino core's LED_BUILTIN

4. Power Supply

The board can be powered from the USB-C connector (5V) or an external 5V supply applied to the 5V pin on J2. Two linear regulators derive the rails THEJAS32 requires:

Ref	Part	Conversion	Function
U2	AMS1117-3.3	5V -> 3.3V	Powers VDDIO (I/O rails), CP2102N, BW16 module, and the 3V3 header pin
U3	AMS1117-1.2	3.3V -> 1.2V	Powers VDD (THEJAS32 core logic voltage)

Each rail is decoupled per the THEJAS32 datasheet's recommended scheme (0.1µF / 0.01µF / 10µF ceramic capacitors distributed across VDD and VDDIO pins).

5. Boot Modes & Programming

Mode	Jumper	Behaviour
Serial / UART (default)	JP1 open	BOOT_SEL = high. Program is transferred over UART0 using the XMODEM protocol via the USB-C / CP2102N link — this is the mode used by the Arduino IDE uploader.
SPI Flash (standalone)	JP1 shorted	BOOT_SEL = low. On every power-up, the bootloader copies the first 250 KB from the onboard W25Q16JVSS flash into SRAM and jumps to it — no PC connection needed.

Note: direct upload-to-SPI-flash is not yet implemented in the v1.0 SDK; SPI-flash mode currently requires flashing the part by other means before use.

5.1 Programming from the Arduino IDE

- Open Arduino IDE -> File -> Preferences -> Additional Boards Manager URLs, and add: https://raw.githubusercontent.com/nishil-26/arc32-sdk/main/package_arc32_index.json
- Open Boards Manager, install "Arc32 RISC-V Boards" — this also downloads the riscv-none-elf-gcc toolchain (xPack, v15.2.0-1) for your OS automatically
- Select Tools -> Board -> Arc32 (THEJAS32 RISC-V), choose the correct COM/serial port, and select the upload mode (Serial/UART or SPI Flash) from Tools -> Upload Mode
- Compile and upload as with any Arduino board — max sketch size 262,144 bytes (256 KB), matching the onboard SRAM

6. Electrical Characteristics

These values are inherited directly from the THEJAS32 SoC datasheet (C-DAC / VEGA Processors, Doc Ver 1.0) and apply to the SoC as used on the Arc32 v1 NEO board.

6.1 Absolute Maximum Ratings

Parameter	Rating
VDD (core)	1.5 V max
VDDIO (I/O)	3.9 V max
Operating temperature	0 to +70 °C
I/O source/sink current (PROC_HB, GPIO0/1, SPI0/1/3)	12 mA
I/O source/sink current (I2C0-2, PWM, UART0-2)	8 mA
ESD (HBM / CDM / MM)	2000 V / 500 V / 200 V

6.2 General Operating Conditions

Symbol	Min	Nom	Max
VDD (core, 1.2V rail)	1.08 V	1.2 V	1.32 V
VDDIO (I/O, 3.3V rail)	2.97 V	3.3 V	3.63 V
Operating temperature	0 °C	+25 °C	+70 °C

6.3 I/O Characteristics

Parameter	Min	Max
VIH — High-level input voltage	2 V	VDDIO + 0.3 V
VIL — Low-level input voltage	—	0.4 V
VOH — High-level output voltage	2.4 V	—
VOL — Low-level output voltage	—	0.8 V

Note: THEJAS32 GPIOs have no internal pull-up/down resistors — use an external ~10 kΩ resistor where a pull-up is required (e.g. I2C bus lines, INPUT_PULLUP mode).

7. Mechanical

Parameter	Value
Board outline	60.42 mm x 89.83 mm (from Edge_Cuts layer)
Header pitch	2.54 mm (0.1 in), single row
Header pin counts	J2: 20, J3: 20, J4: 12
SoC package	THEJAS32, LQFP128, 14 x 14 mm
USB connector	USB Type-C receptacle

8. Example Sketches

The Arc32_Examples library (bundled with the SDK) includes ready-to-run sketches:

- Blink — toggles the onboard User LED (LED_BUILTIN / D19-D20 / GPIO22)
- SerialEcho — echoes characters back over the USB Serial Monitor
- I2C_Scanner — scans the I2C0 bus (J3 SCL/SDA) for connected devices

9. Known Limitations (SDK v1.0)

- No C++ STL (<algorithm>, std::vector, std::function, ...) — the bare-metal RV32IM toolchain has no libstdc++ port; only core C++ + the bundled Print/Stream/String classes are available
- No hardware FPU — all float/double math runs in software via libgcc soft-float (slower but functional)
- No analogRead() / ADC — THEJAS32 has no built-in ADC; use an external I2C ADC (e.g. ADS1115) over I2C0/I2C1 via the Wire library until a dedicated wrapper ships
- Direct sketch upload to SPI flash is not yet implemented
- INPUT_PULLUP has no internal pull-up hardware on THEJAS32 GPIOs and is currently treated as plain INPUT — add an external ~10 kΩ pull-up resistor
- Wi-Fi / Bluetooth (BW16) driver support is in active development and not yet exposed as an Arduino library

10. Revision History

Revision	Date	Changes
Rev 1.0	July 2026	Initial public release of the Arc32 v1 NEO datasheet

This board is an independent, community-designed product built around the THEJAS32 SoC (C-DAC / VEGA Processors) and is not an official C-DAC or VEGA Processors product. Designed & maintained by Nishil Patel as a part of project. Hardware design, SDK source, issue tracker and licensing terms: github.com/nishil-26/arc32-sdk.